

A Hierarchical ReAct Multi-Agent Pipeline for Residential Appliance Load Scheduling Under Dynamic Electricity Pricing

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Abstract

Households on dynamic electricity contracts can lower their bills by running flexible appliances when electricity is cheap, but coordinating several appliances under a volatile price curve and a household peak load limit is difficult to do by hand. This thesis follows Design Science Research to design and evaluate a hierarchical ReAct multi-agent pipeline, built in LangGraph, that gives a household a daily appliance recommendation. An Orchestrator, an Appliance Profiler, a Price-Aware Scheduler and a Conflict Resolver work from the day's appliance plan, the base load, the peak limit and the prices, and a verification step returns flawed schedules for revision. The study asks how much the recommendations reduce cost compared with no scheduling, a language model without tools, rule-based automations and the open-source optimizer EMHASS (SQ1), and what the multi-agent design changes compared with a single agent (SQ2).

The evaluation combines appliance cycles from the REFIT dataset with Finnish Nord Pool prices in 48 scenarios, scores every schedule against a MILP optimum, and confirms the main hypotheses on 48 held-out scenarios with fifteen-minute prices after pre-registration. With GPT-4o-mini, both agent designs come within about one percent of the optimal cost and capture about 90 % of the achievable saving, whereas the same model without tools captures 13 to 24 %. The pipeline and the single agent are statistically equivalent in cost, and the pipeline uses about 30 % fewer input tokens at a longer response time. A simple rule-based timer schedules slightly better than both agents, and EMHASS configured with measured power profiles is not significantly different from them, while the agents outperform EMHASS with nominal power and a common cheapest-hours automation, which exceeds the peak limit on about half of the days. Tool use is thus essential, and any advantage of the agents lies in their explanations and potential flexibility rather than in cost. The comparison with a small model run locally is in progress.

Keywords

multi-agent system, ReAct, LangGraph, home energy management, appliance scheduling, dynamic electricity pricing, Design Science Research

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Foreword

Work in progress.

Abbreviations

API	Application Programming Interface
DSR	Design Science Research
HEMS	Home Energy Management System
LLM	Large Language Model
MAS	Multi-Agent System
MCP	Model Context Protocol
MILP	Mixed Integer Linear Programming
Nord Pool FI	Nord Pool Finland Bidding Zone
ReAct	Reasoning and Acting
REFIT	Personalised Retrofit Decision Support Tools for UK Homes using Smart Home Technology (the research project that produced the dataset)
SLM	Small Language Model
SLR	Systematic Literature Review

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1. Introduction

Driven by dynamic pricing and smart grid infrastructure, an architectural shift is transforming how residential electricity consumption is monitored, scheduled, and optimized. The proliferation of smart metering infrastructure, the growing adoption of dynamic electricity pricing contracts, and the expanding penetration of renewable energy sources are together creating both an opportunity and a challenge for residential energy management. Households with dynamic spot-price contracts can substantially reduce their electricity bills by shifting flexible appliance loads to cheap off-peak windows. However, the cognitive burden of coordinating multiple appliance cycles across a volatile 24-hour price curve exceeds what most households can practically manage without automated assistance.

In Finland, this challenge has reached particular salience. Since October 2025, Finnish electricity consumers on spot-price contracts face 15-minute pricing intervals tied to the Nordic spot market (Nord Pool FI bidding zone), and prices within a day vary widely. From 2023 to 2026, the most expensive hour of a day cost a median of 3.1 times the cheapest on days without near-zero prices, and one day in five had at least one hour with a negative price (computed from Nord Pool FI prices; Pakastin Oy, 2026). Such extreme prices measurably change household consumption and affect the distribution network (Haakana et al., 2025). At the same time, research in LLM-based Home Energy Management Systems (HEMS) has advanced rapidly. Recent hierarchical multi-agent implementations demonstrate that language model reasoning can be applied to appliance scheduling tasks that previously required domain-expert rule encoding or computationally expensive optimization solvers (El Makroum et al., 2025; Jonah et al., 2026). However, these early implementations have documented limitations that this thesis addresses.

1.1 Background and motivation

The research journey toward optimizing residential energy consumption is shifting from rigid heuristic algorithms to autonomous, reasoning-capable AI architectures. Conventional HEMS solutions have historically relied on mathematical optimization solvers, rule-based schedulers, or classical machine learning models trained on historical consumption patterns (Beaudin & Zareipour, 2015). While effective in constrained and well-defined settings, these approaches require significant domain expertise to configure and do not generalize to novel price regimes.

The emergence of Large Language Models (LLMs) as general-purpose reasoning engines has opened new possibilities for HEMS design. Yao et al. (2023) introduce the ReAct paradigm, a Thought-Action-Observation loop that interleaves reasoning with tool calls. They demonstrate that LLMs equipped with this loop can solve multi-step planning problems requiring dynamic information retrieval and constraint satisfaction. Applied to

residential scheduling, a ReAct agent can reason over a price forecast, classify appliances by shiftability, identify the cheapest available time window, and verify constraint compliance through a traceable reasoning chain.

A secondary motivation concerns sustainable AI deployment. The energy needed per inference differs by orders of magnitude between models and tasks (Luccioni et al., 2024), and serving a small language model (SLM) of about 7 billion parameters has been estimated to be 10 to 30 times cheaper in latency, energy and computation than serving a model of 70 to 175 billion parameters (Belcak et al., 2025). Whether such smaller models are good enough for a given agentic task is an empirical question. Since the Low Carbon Alert trainee project context prioritizes energy efficiency in the AI systems used to schedule household loads, a secondary evaluation axis compares the commercial cloud model GPT-4o-mini with gpt-oss-20b, an open-weight mixture-of-experts model (about 21 billion parameters, of which about 3.6 billion are active per token) run locally on a laptop, contributing to the research agenda on sustainable agentic AI deployment.

1.2 Research problem

Existing LLM-based HEMS implementations have documented limitations. In the hierarchical multi-agent system of El Makroum et al. (2025), only one of three evaluated models coordinated all appliances, and the evaluation covers a single day. Reviewing that work, Jonah et al. (2026) list multi-appliance coordination failures, scheduling on price alone, a single-day evaluation horizon and untested constraint safety as open problems. Their own single-agent system is evaluated for one household over a few weeks, and they recommend that every schedule be validated independently before it is executed. Robustness across price-volatility regimes and a controlled comparison of multi-agent and single-agent designs therefore remain open. In addition, it remains unclear what a language model agent adds over simpler approaches that households can already use, such as rule-based automations and open-source optimizers.

The artifact studied in this thesis gives each household a daily recommendation. Every day it receives the household's appliance plan for that day, namely which appliance cycles are to run and when each is ready and must be finished, together with the household's base load, its peak load limit and the day's electricity prices. It returns a start time for each appliance and a short explanation. To address the limitations above, the study is guided by one main research question and two sub-questions:

> **RQ:** How can a hierarchical ReAct multi-agent pipeline be designed to give households daily appliance recommendations under dynamic electricity prices, and how much do its recommendations reduce electricity cost compared with single-agent, rule-based and optimal scheduling?
>
> **SQ1:** How much do the pipeline's daily recommendations reduce electricity cost compared with no scheduling, a language model without tools, rule-based scheduling and a deployed home energy management optimizer?

> **SQ2:** Does splitting the task among specialist agents change the quality, inference cost and latency of the recommendations compared with a single agent, and does this still hold when a small model run locally replaces the commercial cloud model?

The main RQ combines the design of the artifact with a measure of what its recommendations are worth. SQ1 places the pipeline among the alternatives a household has: doing nothing, asking a language model directly, a rule-based automation, or a configured optimizer. It also tests the premise of the ReAct approach, that reasoning grounded in tool observations outperforms unaided generation. SQ2 isolates what the multi-agent design itself changes compared with a single agent with the same tools and instructions, and whether the answer depends on the language model. The latter matters for sustainable deployment, because a small local model avoids both API cost and the dependence on a cloud provider. How the results vary with price volatility and appliance type is reported for both sub-questions.

1.3 Design Science Research approach

This study follows the Design Science Research (DSR) methodology as defined by Peffers et al. (2007). DSR is appropriate because the central activity is the iterative construction and empirical validation of a software artifact, namely the hierarchical ReAct multi-agent pipeline, rather than the observation of a pre-existing phenomenon. As Hevner et al. (2004) explain, DSR produces knowledge and understanding of a design problem and its solution through the building and evaluation of an artifact. The artifact is the primary research contribution. A Systematic Literature Review (SLR) is conducted as DSR Phase 1 to confirm the research gap and ground all architectural decisions in prior work.

1.4 Structure of the thesis

This thesis is organized into seven chapters. Chapter 2 presents the literature review covering HEMS concepts, LLM-based agents for energy management, the ReAct framework, multi-agent LLM systems, current agentic AI trends, and the Finnish energy market context. Chapter 3 describes the Design Science Research methodology and evaluation strategy. Chapter 4 presents the system design and implementation. Chapter 5 presents evaluation results across 48 scenarios. Chapter 6 discusses findings, limitations, and future work. Chapter 7 concludes the thesis.

2. Literature Review

This chapter reviews the theoretical background and prior empirical work relevant to the design and evaluation of a hierarchical ReAct multi-agent pipeline for residential appliance scheduling. The review is organized into seven thematic areas. The chapter concludes by synthesizing the literature into a confirmed research gap that motivates the research questions of this thesis.

2.1 Home energy management systems

A Home Energy Management System (HEMS) is a software or hardware system that monitors, controls, and optimizes residential electricity consumption to reduce cost, support grid stability, and improve energy efficiency (Beaudin & Zareipour, 2015). The core scheduling problem is the assignment of flexible appliance loads to time windows where electricity prices are lowest, while ensuring that total household load at any moment does not exceed safe operational limits. A load is flexible when its start time can be shifted without violating user preferences or physical constraints.

HEMS scheduling approaches fall into three broad groups. Rule-based systems apply expert-defined heuristics, for example starting an appliance when the price falls below a threshold. Mathematical optimization methods such as Mixed Integer Linear Programming (MILP) compute optimal schedules for a precisely specified problem; Moradzadeh and Tomsovic (2013), for instance, formulate residential energy management as a two-stage optimization that respects the operational constraints of the distribution network. Such methods require every constraint to be expressed in mathematical form, so a household's preferences must first be translated into that form. Learning-based methods acquire a scheduling policy from data, as in the deep reinforcement learning approach of Mocanu et al. (2019), which is trained on measured building data and then optimizes schedules on-line. None of the three groups explains its decisions in terms that non-expert household users can readily understand, and adapting any of them to a new type of constraint requires a new rule, a new formulation or new training.

Rule-based and optimization approaches are both available to households today as open-source tools. Among Home Assistant users in Finland and the other Nordic countries, a widely shared approach is to automate devices with community blueprints that switch a device on during the cheapest hours of a user-defined time window, based on Nord Pool day-ahead prices (Kert, n.d.). Such rules handle each device separately and do not consider the household's total load. EMHASS (Energy Management for Home Assistant) is an open-source optimizer that schedules deferrable loads, batteries and photovoltaic production against price forecasts with linear and mixed-integer programming (EMHASS contributors, 2026). It supports a time window for each load, a limit on the power drawn from the grid, and loads defined by a fixed power profile. Both

kinds of tool require the household to describe each appliance in structured form, namely its power, run time and time window. This thesis uses both as reference points for the agents (Section 3.2).

2.2 LLM-based agents for energy management

The application of LLMs to energy management tasks has emerged as an active research area since 2023. El Makroum et al. (2025) present a hierarchical agentic AI architecture with one orchestrator agent and three specialist agents that follow the ReAct pattern, evaluated with three open-weight models (Llama 3.3 70B, Qwen 3 32B and GPT-OSS 120B) on Austrian day-ahead prices against MILP benchmarks. All three models scheduled single appliances optimally, but only Llama 3.3 70B coordinated all appliances in the multi-appliance scenarios. The evaluation comprises 75 runs on a single day for one household with a washing machine, a dishwasher and an electric vehicle charger, and the authors note that thermal loads such as heat pumps, electric heating and hot water storage are not included. Of the published systems, this design is the closest to the pipeline of this thesis, but it is not compared with a single agent and the price conditions are not varied.

Jonah et al. (2026) evaluate a single tool-calling ReAct agent with half-hourly dynamic prices from the United Kingdom, weather and photovoltaic forecasts and household demand data, using five language models, among them GPT-4o-mini. Building on El Makroum et al., they list multi-appliance coordination failures, scheduling on price alone, a single-day evaluation horizon and untested constraint safety as open problems. Their agents capture 96.7 to 98.0 % of the savings available between an off-peak timer and a MILP optimum, but the authors also observe fabricated schedules and cases in which a model reasons correctly about a constraint and still commits a schedule that violates it. They therefore recommend that every schedule be validated independently against duration, deadline and power constraints before it is executed. Their evaluation covers one household over twelve tariff days and a one-week deployment. These open problems shape the design of this thesis: coordination under a household peak load limit, a deterministic verification step for every schedule, a controlled comparison with a single agent, and an evaluation across six price-volatility regimes on two benchmarks.

2.3 The ReAct framework

The ReAct (Reasoning and Acting) framework, introduced by Yao et al. (2023), proposes that LLM agents should interleave reasoning traces with action execution rather than separating reasoning and acting into sequential phases. In the ReAct paradigm, each agent step produces three outputs. The Thought is explicit reasoning about the current state and the next action, the Action is a structured call to an external tool or API, and the Observation is the result returned by that tool. This loop repeats until the agent reaches a final answer.

Yao et al. (2023) evaluate ReAct against chain-of-thought reasoning and direct action baselines across four benchmark tasks. ReAct outperforms both baselines on all four tasks, with particularly large gains on tasks requiring dynamic information retrieval and multi-step planning. These are precisely the characteristics that distinguish HEMS scheduling from simpler single-step optimization tasks. The interpretability of the ReAct trace is a distinctive advantage for energy management applications: because each agent step records the reasoning behind every tool call and schedule decision, the full trace provides an auditable explanation of why a particular appliance was assigned to a particular time window.

2.4 Multi-agent LLM systems

Multi-agent LLM systems have been proposed as a solution to the context window and coordination limitations of single-agent designs. Han et al. (2024) discuss the challenges and open problems of LLM multi-agent systems, among them task allocation, robust reasoning through iterative debate, the management of complex and layered context information, and memory management. Role specialization, in which each agent is responsible for one well-defined subtask and has access only to the tools that subtask needs, is one response to these challenges and the design principle followed in this thesis.

Hong et al. (2024) demonstrate role-based multi-agent orchestration for software development with MetaGPT, in which agents act as product manager, architect, project manager, engineer and quality-assurance engineer. Encoding standard operating procedures and requiring structured intermediate outputs, exchanged through a shared message pool, reduces the inconsistencies that arise when language model calls are simply chained, and MetaGPT reported state-of-the-art results on code-generation benchmarks at the time. Structured hand-offs between specialists are directly relevant to the multi-appliance coordination failures documented in LLM-based HEMS (El Makroum et al., 2025).

2.5 Agentic AI trends in 2025–2026

Several 2025–2026 trends in agentic AI are directly applicable to this thesis. The most significant is the shift from single agents to systems in which several LLM-based agents with distinct roles cooperate, a research area that has grown rapidly since 2023 (Guo et al., 2024; Wang et al., 2024). These surveys describe how the profiles, memory, planning and communication of such agents are organized, and they identify coordination between agents and the evaluation of multi-agent systems as open challenges. The motivation is that a complex task can be divided among specialists, each with a smaller context and fewer tools than a single agent that must handle the whole task.

The Model Context Protocol (MCP), introduced by Anthropic in November 2024 and donated to the Linux Foundation’s Agentic AI Foundation in December 2025, has

emerged as the industry standard for agent-to-tool communication (Anthropic, 2024). MCP provides a standardized, model-agnostic interface that enables AI agents to connect to external tools and data sources, eliminating the brittle custom integration code that previously characterized agent tool implementations. The LangChain MCP adapters library makes MCP tools directly compatible with LangGraph agents, enabling the price API and data reader in this thesis to be implemented as reusable MCP servers.

Liu et al. (2025) catalogue eighteen architectural patterns for agents built on foundation models. Three of them correspond directly to designs in this thesis: role-based cooperation, in which agents with assigned roles divide the work, describes the multi-agent pipeline; one-shot model querying, in which the model produces its answer in a single step, describes the LLM-without-tools baseline; and human reflection, in which humans give feedback on the agent's plan, describes a human-in-the-loop (HITL) checkpoint. Guidelines for human-AI interaction likewise recommend that a system make clear why it did what it did and support efficient correction by the user (Amershi et al., 2019). In this thesis, a lightweight HITL checkpoint where the Orchestrator presents the proposed schedule to the user for approval before the Conflict Resolver finalizes it is incorporated as an optional LangGraph interrupt node, aligning the design with current best practice for high-stakes autonomous decisions. The checkpoint is available in the artifact but was not enabled in the reported evaluations, which run without human intervention.

2.6 Finnish energy market context

Finland provides a directly relevant operational context for this research. The country operates within the Nordic spot electricity market (Nord Pool) and as of October 2025 has adopted 15-minute pricing intervals, making it one of the most granularly priced residential electricity markets in Europe. At the end of 2024, about 33 % of all Finnish electricity contracts were dynamic-price contracts, up from 9 % in 2021, and these expose consumers directly to hourly and quarter-hourly price fluctuations (Energy Authority, 2025).

Intraday price volatility in the Finnish market is substantial. In the Nord Pool FI day-ahead prices from 2023 to 2026, the difference between the cheapest and the most expensive hour of a day had a median of 5.7 c/kWh excluding VAT and exceeded 10 c/kWh on 28 % of days. On days without near-zero prices the most expensive hour cost a median of 3.1 times the cheapest, and on 19 % of days at least one hour had a negative price (computed from Pakastin Oy, 2026). Extremes are far larger: the highest hourly price in the period, 189.6 c/kWh on 5 January 2024 at 19:00, was about one hundred times the median night-time price of 1.8 c/kWh. On 24 November 2023 the price fell to -500 EUR/MWh for half a day, and households with spot-price contracts noticeably increased their consumption (Haakana et al., 2025). At the same time, an estimated 7 to 15 % of Finnish households are energy poor (Aalto University, 2026). These structural features make Finnish residential demand flexibility not merely a consumer cost issue but a national energy system challenge.

2.7 Literature synthesis and research gap

The literature reviewed across the preceding sections establishes three converging findings. First, HEMS scheduling is a multi-step, multi-constraint optimization problem that LLM-based agents can address more flexibly than rule-based or classical optimization approaches. Second, the first LLM-based HEMS implementations, both hierarchical and single-agent, have documented limitations in multi-appliance coordination, in scheduling on price alone, in the safety of committed schedules, and in evaluation scope (El Makroum et al., 2025; Jonah et al., 2026). Third, multi-agent role-specialized designs are reported to outperform single-agent baselines on complex multi-step tasks in other domains (Hong et al., 2024), but no published HEMS study compares such a design with a single agent under controlled conditions across multiple price-volatility regimes.

This gap is the direct motivation for this thesis. The study makes three contributions. The first is a hierarchical ReAct multi-agent pipeline that gives households daily appliance recommendations, implemented in LangGraph. The second is a measure of how much these recommendations reduce electricity cost compared with no scheduling, a language model without tools, rule-based automations, a deployed open-source optimizer and the mathematical optimum, spanning six price-volatility regimes (SQ1). The third is a controlled comparison of the multi-agent and single-agent designs in recommendation quality, inference cost and latency, with the commercial GPT-4o-mini and the locally run open-weight gpt-oss-20b (SQ2). The hypotheses for the comparisons with GPT-4o-mini were fixed in advance and confirmed on a held-out benchmark.

3. Research Methodology

This chapter describes the research methodology and explains how Design Science Research is applied in this thesis. It presents the rationale for the selected approach, the phase mapping to thesis activities, and the evaluation strategy including the statistical methods used.

3.1 Design Science Research methodology

This thesis follows the Design Science Research (DSR) methodology as defined by Peffers et al. (2007). DSR is a research paradigm focused on the design and evaluation of innovative artifacts that solve complex practical problems. As Hevner et al. (2004) argue, design science creates and evaluates IT artifacts intended to solve identified organizational problems, producing knowledge through the construction and evaluation of artifacts rather than through the observation of natural phenomena. DSR is the appropriate methodology for this thesis because the central activity is building and evaluating the multi-agent pipeline artifact, not observing a pre-existing system.

Artifacts in DSR are classified into four types: constructs, models, methods, and instantiations (March & Smith, 1995). In this thesis, the artifact is an instantiation, that is, a working software implementation of a hierarchical multi-agent scheduling pipeline. It is supported by architectural models (the LangGraph state graph design) and constructs (the agent role taxonomy: Profiler, Scheduler, Resolver, Orchestrator).

Table 3.1 maps the six DSR phases defined by Peffers et al. (2007) to the specific activities and outputs in this thesis.

Table 3.1 DSR phase mapping.

Phase	DSR step	Activity in this thesis	Output
1	Problem identification	SLR confirming no multi-agent ReAct HEMS addresses multi-appliance coordination across multiple price-volatility regimes	Gap statement; confirmed RQs
2	Define objectives	Specify measurable success criteria: optimality gap target, conflict resolution rate, latency, token cost	Objective specification
3	Design and development	Implement Appliance Profiler, Price-Aware Scheduler, Conflict Resolver, Orchestrator, and single-agent baseline in LangGraph	Working pipeline artifact
4	Demonstration	Run pipeline on five representative scenarios; confirm valid cost-reducing schedules	Demonstration results
5	Evaluation	Exploratory run on 48 hourly-price scenarios; pre-registered confirmatory run on 48 held-out fifteen-minute scenarios; comparison with rule-based and real home energy management tools; Wilcoxon and TOST tests	Quantitative evaluation results
6	Communication	Write findings, design guidelines, limitations, and future work	Complete thesis

3.2 Evaluation strategy

The evaluation follows a simulation-based controlled design. No real sensors, buildings, or hardware are required. All inputs are drawn from publicly available datasets and APIs, and all evaluation scenarios are constructed from historical real-world data rather than synthetic simulations. The evaluation comparison is paired: the same 48 scenarios are run through the multi-agent pipeline and the single-agent baseline with each of the two language models, and through four further schedulers. The MILP solver provides the optimal schedule, and a naive schedule starts every appliance as soon as the user allows. Two baselines test whether the agents add value over simpler approaches. A rule-based smart timer, of the kind found in commercial smart plugs and home energy applications, places the appliances one at a time, the most energy-intensive first, each at its cheapest start at which the household load stays within the peak limit. It needs structured appliance data but no language model. The rule was fixed before it was run on the benchmark and was not tuned afterwards. The LLM-without-tools baseline uses the same model as the agents, GPT-4o-mini, but gives it the day's appliance plan together with all household data and the price and measured base load of every slot in a single prompt, and takes its schedule from one answer without tools or a reasoning loop. This corresponds to the standard prompting baseline of Yao et al. (2023). Because the model is the same, the comparison with it isolates the contribution of tool use; it is run with GPT-4o-mini only.

All three systems that use a language model (the pipeline, the single agent and the LLM without tools) thus run on the same model and differ only in how it is used. Because language model output varies between runs, each system that uses a language model is run three times per scenario and model, while the deterministic schedulers are run once.

After the pre-registration, two real home energy management tools were added as secondary baselines, and their results are reported descriptively. EMHASS 0.18.4 is run as households run it: as a Docker service, with its default settings, called through its REST interface. Each scenario is sent as runtime parameters of a day-ahead optimization: the prices, the base load as the load forecast, no photovoltaic production, the appliances as deferrable loads with their time windows, and the peak limit as the maximum power drawn from the grid. Home Assistant itself is not involved, since all data is passed with the request. EMHASS is run in two configurations: with the measured power profile of each appliance cycle, and with the simpler configuration of a constant nominal power, set to the cycle's peak, for the cycle's duration. As a cross-check, the same scenarios were also solved by calling EMHASS's optimization module directly with an exact solver tolerance; the two routes give the same schedule in most scenarios, and the remaining differences stem from the default tolerance of the service. The cheapest-hours automation reproduces the selection rule of the Home Assistant blueprint by Kert (n.d.). Within the appliance's window, the prices are sorted in ascending order and the threshold is the price at index $4h$ (counting from zero), where h is the cycle length rounded up to whole hours. The blueprint switches a device on whenever the price is at or below the threshold; for a cycle appliance, it is assumed that this trigger starts the cycle, which then runs to completion. Each appliance is handled by its own automation.

The evaluation proceeds in two stages. The main benchmark of 48 scenarios with hourly prices (Section 5.1) is the exploratory stage. It was used while the pipeline was refined over several design iterations, and the hypotheses for SQ1 and SQ2 were formulated after its results had been inspected, so its results are reported as exploratory. The hypotheses were then fixed in a pre-registration, committed to the project repository before a confirmatory run on a second benchmark of 48 scenarios built from fifteen-minute prices (Section 5.2), on which no language model had previously been run. Only the confirmatory run is used to accept or reject H1 and H2. The comparison of the two language models (H3) was part of the original research plan and is tested on the main benchmark; the direction of its hypothesis was stated after the pilot runs, which had indicated it. The research questions were consolidated from four sub-questions into two after the pre-registration; the registered hypotheses and tests were not changed, and the pre-registration records this as a dated amendment. Table 3.2 lists the hypotheses and their tests.

3.2.1 Scoring

All schedules are scored by the same function. The cost of a schedule is the consumer price of electricity multiplied by the energy of the flexible appliance cycles, summed over the day; the consumer price adds the retail margin, distribution fee, electricity tax and VAT to the spot price (Section 5.1). The base load costs the same under every schedule and is

Table 3.2 Hypotheses and statistical tests.

	Hypothesis	Test
H1	Both agent designs have a lower optimality gap than the LLM without tools (SQ1).	Wilcoxon signed-rank, per design
H2a	The pipeline and the single agent are equivalent in optimality gap within ± 1 percentage point (SQ2).	Paired t TOST
H2b	The pipeline uses fewer input tokens and has a lower API cost per run than the single agent (SQ2).	Wilcoxon signed-rank
H2c	The pipeline has a longer latency than the single agent (SQ2).	Wilcoxon signed-rank
H3	With gpt-oss-20b instead of GPT-4o-mini, the feasibility rate of the pipeline falls more than that of the single agent (SQ2).	Wilcoxon signed-rank on the per-scenario change

therefore excluded from the cost, but it counts towards the peak load limit. A schedule is feasible when every appliance has a start time, every cycle runs within its user window, and the total household load never exceeds the peak limit. An infeasible schedule, or a run that returns no schedule, is scored at the cost of the naive schedule, on the reasoning that the household would then fall back to starting its appliances when they are ready. The feasibility rate is reported separately, so that this convention does not hide failures.

The primary evaluation metric is the schedule optimality gap:

$$\text{optimality gap}(\%) = \frac{\text{agent cost} - \text{MILP cost}}{\text{MILP cost}} \times 100.$$

A smaller optimality gap indicates a schedule closer to the mathematical optimum. Because the consumer price stays positive even when the spot price is negative, the MILP cost is always positive and the gap is well defined. As a complementary measure, the share of the achievable saving that a system captures over a set of runs is

$$\text{savings captured} = \frac{\sum \text{naive cost} - \sum \text{agent cost}}{\sum \text{naive cost} - \sum \text{MILP cost}},$$

where the sums run over all runs of the system. It equals one for optimal schedules and zero for schedules no better than the naive ones. The ratio is pooled over the runs rather than averaged per scenario, because on some days the naive schedule is already almost optimal: with an achievable saving of a fraction of a cent, a schedule that costs one cent more would count as minus several thousand percent and dominate the average.

The choice of tariff does not affect the comparison between systems. The margin, distribution fee and electricity tax are constant per kilowatt-hour, and every cycle uses the same energy wherever it is placed, so the consumer price changes the cost of every schedule by the same amount and scales all cost differences by the common factor $1 + v$. The optimal schedule, the ranking of the systems and the savings captured are therefore identical to those under the spot price alone, which was also verified on the recorded

schedules. Only the optimality gap depends on the tariff, through its denominator: the spot cost of the optimal schedule is a median of 21 % of its consumer cost on the main benchmark and 11 % on the held-out benchmark, so gaps measured against the spot cost alone would be several times larger. The consumer price is used because it is what the household pays, and because it keeps the denominator positive on negative-price days.

The sub-questions use these metrics as follows. SQ1 compares the optimality gap, feasibility and savings captured of the pipeline's recommendations with those of no scheduling, the LLM without tools, the rule-based schedulers and EMHASS. SQ2 compares the optimality gap, input and output tokens, API cost and latency of the pipeline and the single agent, and the change in feasibility rate, optimality gap, latency and token use when GPT-4o-mini is replaced by gpt-oss-20b. Two secondary analyses are reported without hypotheses: the conflict resolution rate, which is the share of feasible runs among the scenarios that require coordination (those in which starting every appliance at its individually cheapest time would exceed the peak limit), and the relation between the saving of each appliance cycle and the cycle's duration, power draw, energy and window width.

3.2.2 Statistical analysis

The multi-agent and single-agent designs are compared per scenario, using the mean optimality gap over the three runs as the paired observation. The Wilcoxon signed-rank test assesses whether the difference across the 48 paired scenarios is statistically significant. The test is appropriate because it is non-parametric, requires no assumption of normality, and is designed for paired comparisons. Pratt's treatment of zero differences is used, so that scenarios in which both designs reach the same cost remain in the ranking instead of being discarded. Statistical significance is assessed at $p < 0.05$ two-tailed, and practical significance is reported with the effect size $r = |Z|/\sqrt{N}$. The test is repeated per scenario type to identify the price-volatility conditions under which the difference is most pronounced. The relation between appliance characteristics and savings is assessed with Spearman rank correlation. All tests are two-sided. A directional hypothesis is supported when its test is significant and the effect has the predicted direction. Two-sided tests are the conservative choice for directional hypotheses, since a one-sided test would halve each p-value. The p-values of the registered hypotheses are adjusted for multiple testing with the Holm procedure within two families: H1 with two tests (one per agent design), and H2 with four tests (H2a, H2b for input tokens, H2b for API cost, and H2c).

A non-significant difference does not by itself show that the two designs perform alike, since it can also result from too little data. Equivalence is therefore tested separately with the two one-sided tests (TOST) procedure (Schuirmann, 1987; Lakens, 2017). The designs are considered equivalent when both one-sided tests reject, at the 5 % level, a mean difference in optimality gap of at least one margin in either direction. This is the same as requiring the 90 % confidence interval of the mean paired difference to lie inside the margin. The primary margin is ± 1 percentage point. The optimal daily cost of the flexible cycles has a median of € 0.46 on the main benchmark (interquartile range € 0.30

to €1.48, maximum €2.92) and €0.50 on the held-out benchmark (maximum €4.48). One percentage point therefore amounts to about half a cent on the median day and at most three to four and a half cents on the most expensive day, which a household would not notice on its bill. The paired t version of the test concerns the mean difference and is the primary one. A Wilcoxon version, which concerns the typical scenario, is reported alongside because the differences are heavy-tailed. The equivalence test and its margin were chosen after the exploratory results had been seen. The margin is therefore fixed in the pre-registration for the confirmatory run, and margins of 0.5 and 2 percentage points are reported alongside to keep the choice transparent.

Efficiency is compared with the same paired Wilcoxon test on the per-scenario means of input tokens, output tokens, API cost, latency and the number of tool calls per run. Each agent design is also compared with the smart timer and the LLM without tools using the paired Wilcoxon test on the optimality gap.

4. System Design and Implementation

This chapter describes the design and implementation of the hierarchical ReAct multi-agent pipeline. It presents the problem definition and design objectives, the architecture of the complete pipeline, the specifications of each agent, the single-agent baseline implementation, the MILP ground truth solver, and the technology stack used throughout.

4.1 Problem definition and objectives

Based on the literature synthesis in Section 2.7 and DSR Phase 2, the artifact must satisfy five design objectives. First, multi-appliance coordination: the pipeline must correctly schedule two or more flexible appliances simultaneously without producing overlapping high-load windows. Second, price-volatility robustness: the pipeline must produce cost-reducing schedules across all six scenario types. Third, constraint compliance: every schedule must respect user-defined completion deadlines and household load limits. Fourth, interpretability: the full reasoning trace of every scheduling decision must be captured through LangSmith. Fifth, computational efficiency: the pipeline must complete scheduling within 60 seconds and under $0.10 API cost per scenario using GPT-4o-mini.

4.2 Architecture overview

The artifact is a LangGraph state graph in which each node is either a ReAct agent or a coordination step. All nodes read from and write to a shared state object that persists for the whole run (Table 4.1). The Orchestrator first turns the household's request into a task brief listing the appliances to schedule. The Appliance Profiler then classifies the appliances and records the power draw, cycle duration and time window of every flexible one. The Price-Aware Scheduler proposes a start time for each flexible appliance from the price curve, and the Conflict Resolver checks the proposal against the household peak load limit and staggers start times where appliances overlap.

The graph is not a single pass. After the Resolver, the Orchestrator's verification step checks the resolved schedule for three kinds of problems: remaining peak load conflicts, detected with the same conflict tool that the Resolver uses; requested appliances that have no start time; and start times outside the user's time window. If any problem remains, the step returns a description of it to the Scheduler, which proposes a revised schedule that the Resolver checks again. The Scheduler and Resolver therefore run at most twice, that is, with one feedback round; the schedule is then finalised, and the Orchestrator writes a short summary for the household that states the result of the final peak load check. The original request travels in the shared state together with the Orchestrator's brief, so an

appliance missing from the brief can still be profiled and scheduled. Figure 4.1 shows the architecture, and Table 4.1 lists the fields of the shared state.

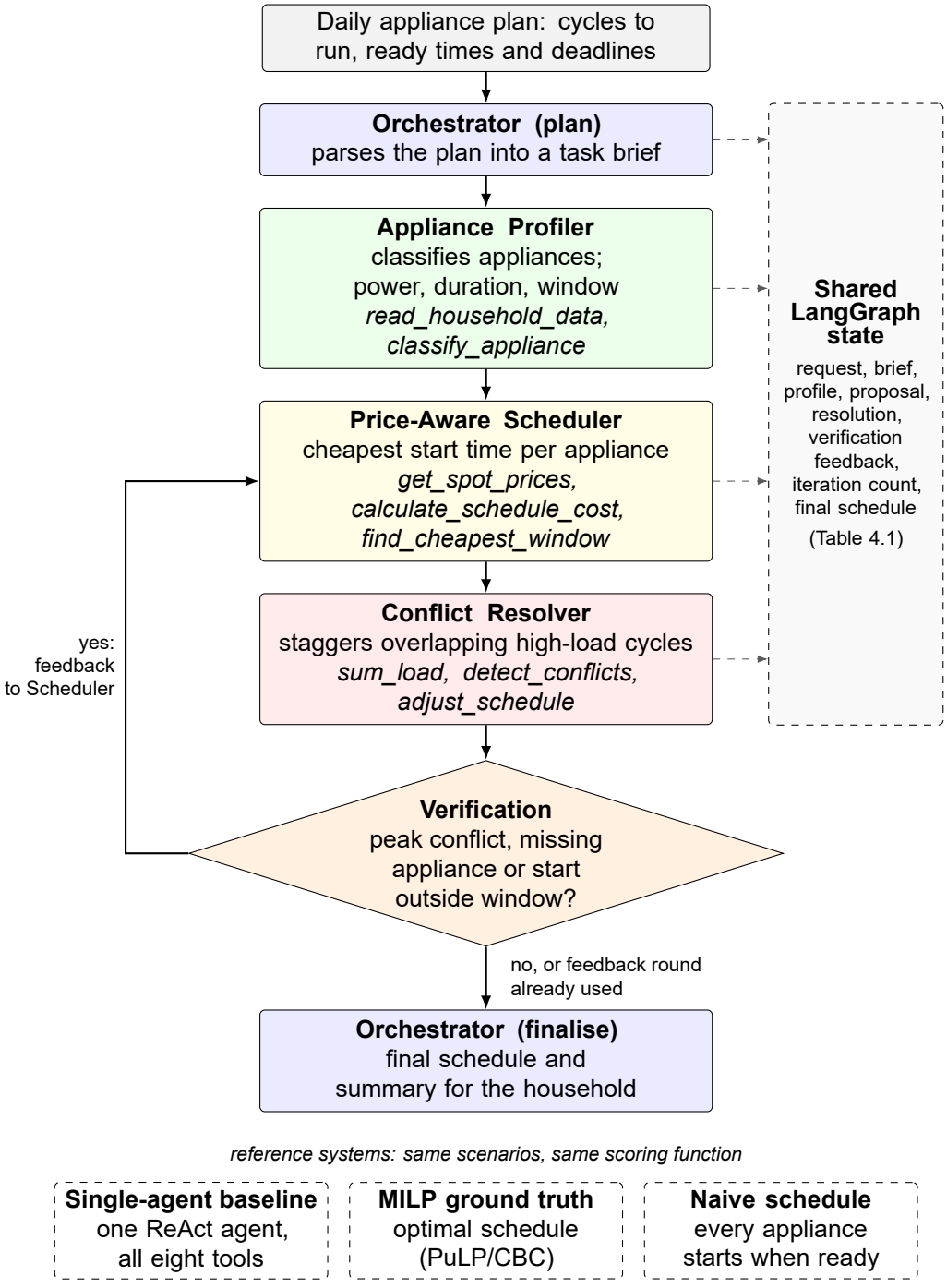


Figure 4.1. Architecture of the hierarchical ReAct multi-agent pipeline.

Table 4.1 Fields of the shared LangGraph state.

Field	Written by	Content
request	caller	the household's scheduling request in natural language
brief	Orchestrator	appliance identifiers to schedule and stated constraints
profile	Profiler	per appliance: type, flexible or fixed, peak power, cycle duration, earliest start and deadline
proposal	Scheduler	start time per appliance, estimated cost and a short justification
resolution	Resolver	conflict-free start times, whether conflicts remain, and the changes made
verification feedback	verification step	remaining peak conflicts, appliances without a start time and start times outside the user window; empty when the schedule passes
iterations	verification step	number of Scheduler–Resolver passes completed (at most two)
final, summary	Orchestrator	final start times and the summary shown to the household
tool calls, recoveries, errors	all nodes	counters and error messages recorded for the evaluation

4.3 Agent specifications

The **Orchestrator** receives the household scheduling request, coordinates the three specialist agents in sequence, and synthesizes the final output. The Orchestrator does not perform scheduling reasoning itself, keeping its context window small and decisions reliable across all scenario types.

The **Appliance Profiler** reads the household consumption data and classifies each appliance by shiftability. The classification distinguishes flexible loads, which have long cycle durations and user-defined completion windows (washing machines, dishwashers and tumble dryers), from fixed loads such as refrigerators and microwaves. The Profiler outputs a structured appliance profile containing power draw in Watts, typical cycle duration in minutes, and the constraint window for each flexible load.

The **Price-Aware Scheduler** receives the appliance profile and the day-ahead price schedule. It reasons through the price series to identify the cheapest available window for each flexible appliance within its constraint envelope and proposes a daily schedule with per-appliance start times and a total cost estimate.

The **Conflict Resolver** receives the proposed schedule and checks for overlapping windows where two or more appliances with high power draw would run simultaneously, exceeding the household's peak load threshold. If conflicts exist, the Resolver proposes

staggered start time adjustments that eliminate the conflict while minimizing the increase in total schedule cost.

Each specialist is a ReAct agent that may only call its own tools and must end with a structured output (the appliance profile, the proposed schedule or the resolved schedule), which is validated against a schema before it is written to the shared state. The agents share a budget of 75 graph steps, divided by workload: 15 for the Profiler, 35 for the Scheduler and 25 for the Resolver. The Scheduler received the largest share after the pilot runs showed that it performs most of the tool calls. When a model replies in prose instead of producing the required structured output, the agent is resumed from its last state and asked once more to return the output, at most twice. This rule applies identically to every agent and model, and every such recovery is recorded.

Table 4.2 lists the eight tools. They are deterministic functions over the scenario data, so every number an agent reasons with comes from a calculation rather than from the language model. No tool solves the whole problem: `find_cheapest_window` considers one appliance at a time and ignores both the other appliances and the peak load limit, so coordinating the appliances is left to the agents.

No tool solves the whole problem. The automatic mode of `adjust_schedule` optimizes the start of one appliance while all others stay where they are; it never re-plans several appliances together. Which appliances to move, in which order, and when to stop remain decisions of the agent, and a poor sequence of single moves can settle on a schedule far from the optimum. In scenario D06, for example, both agent designs ended about 30 % above the optimal cost with a staggered schedule that no single further move could improve.

Table 4.2 Tools available to the agents.

Tool (inputs)	Agent	Output and logic
`read_household_data()`	Profiler	Peak load limit, measured base load range, and for each appliance its type, cycle duration, energy, peak power and user window (flexible) or rated power (fixed).
`classify_appliance(appliance_id)`	Profiler	Flexible or fixed, with a reason; washing machines, dishwashers and tumble dryers are shiftable cycle loads.
`get_spot_prices(from_time, to_time)`	Scheduler	Consumer price in c/kWh for every 15-minute slot, optionally for a time range.
`calculate_schedule_cost(appliance_id, start_time)`	Scheduler	Cost of one cycle started at the given time (power profile $\times$ price $\times$ 0.25 h, summed over the cycle), its end time, and whether it lies within the user window.
`find_cheapest_window(appliance_id, top_k)`	Scheduler	The k cheapest start times within the user window for one appliance, found by evaluating every valid start.
`sum_load(schedule)`	Resolver	Maximum total household load (base load plus scheduled cycles) and the five highest slots, compared with the limit.
`detect_conflicts(schedule)`	Resolver	Time ranges in which the total load exceeds the limit and the appliances running then, or confirmation that there is no conflict.
`adjust_schedule(schedule, appliance_id, new_start_time)`	Resolver	Moves one appliance and reports the cost change, the window check and any remaining overload. Without a start time, it moves the appliance to its cheapest start at which it causes no overload, keeping the others fixed, and lists alternatives.

4.4 Single-agent baseline

The single-agent baseline is one ReAct agent, built with LangChain's `create_agent` on the LangGraph runtime, with access to all eight tools of Table 4.2. It receives the same daily appliance plan as the pipeline, and its system prompt is assembled from the same role instructions that the three specialists receive, so both designs have the same domain knowledge. Its step budget of 75 equals the combined budget of the specialists, and the same structured output schema and recovery rule apply. The only controlled difference between the baseline and the artifact is therefore the presence or absence of role specialization.

The baseline and the pipeline were validated on six pilot scenarios, one per scenario type, before the full evaluation (Section 5.3).

4.5 MILP ground truth solver

The MILP solver computes the mathematically optimal schedule for each scenario. The day is divided into slots $t \in \mathcal{T} = \{0, \ldots, 95\}$ of $\Delta = 0.25\,\text{h}$. Each flexible appliance $a \in \mathcal{A}$ has a power profile $p_{a,0}, \ldots, p_{a,d_a-1}$ in kW over its cycle of d_a slots, an earliest start slot e_a and a deadline slot ℓ_a by which the cycle must end. Its admissible start slots are

$$\mathcal{S}_a = \{\, s \in \mathbb{Z} : e_a \le s \le \ell_a - d_a \,\}.$$

The model uses binary start variables $x_{a,s} \in \{0, 1\}$, equal to one if appliance a starts in slot $s \in \mathcal{S}_a$. With the consumer price π_t in €/kWh, the base load b_t in kW and the household peak limit $\bar{P}$ in kW, the model is

$$\min_{x} \quad \sum_{a \in \mathcal{A}} \sum_{s \in \mathcal{S}_a} c_{a,s}\, x_{a,s}, \qquad c_{a,s} = \Delta \sum_{k=0}^{d_a-1} p_{a,k}\, \pi_{s+k}, \tag{4.1}$$

$$\text{s.t.} \quad \sum_{s \in \mathcal{S}_a} x_{a,s} = 1 \qquad \forall a \in \mathcal{A}, \tag{4.2}$$

$$b_t + \sum_{a \in \mathcal{A}} \sum_{\substack{s \in \mathcal{S}_a \\ 0 \le t-s < d_a}} p_{a,t-s}\, x_{a,s} \le \bar{P} \qquad \forall t \in \mathcal{T}, \tag{4.3}$$

$$x_{a,s} \in \{0, 1\} \qquad \forall a \in \mathcal{A},\ s \in \mathcal{S}_a.$$

The objective (4.1) is the consumer cost of the flexible cycles, with $c_{a,s}$ the precomputed cost of starting appliance a in slot s. Constraint (4.2) requires exactly one cycle per appliance, and constraint (4.3) keeps the total household load within the peak limit in every slot. Because a decision fixes the start of a whole cycle, each cycle runs without interruption once started, and restricting the variables to $\mathcal{S}_a$ keeps every cycle within its user window, so neither property needs a separate constraint. The model is solved with PuLP and the CBC solver in well under a second per scenario. Its correctness was verified against exhaustive enumeration of all start combinations on small scenarios, and it is also used during benchmark construction to confirm that every scenario has a feasible schedule.

4.6 Technology stack

Table 4.3 lists the software used to build and evaluate the artifact, with the versions used for the reported results. All components except the commercial language model are open source. The API cost of the commercial model is an outcome of the evaluation and is reported with the results in Chapter 5.

Table 4.3 Technology stack.

Component	Tool and version	Purpose
Programming language	Python 3.12	implementation of the artifact, baselines and analysis
Agent orchestration	LangGraph 1.2, LangChain 1.4	state graph of the pipeline, ReAct agents, tool calling and schema-validated outputs
Commercial LLM	GPT-4o-mini (snapshot 2024-07-18), OpenAI API	cloud model for both designs
Open-weight LLM	gpt-oss-20b, Ollama 0.34 and 0.35	open-weight model hosted on local hardware (SQ2); Ollama updated itself from 0.34.4 to 0.35.0 during the evaluation
MILP solver	PuLP 3.3 with CBC	optimal reference schedules
HEMS optimizer	EMHASS 0.18.4, Docker	open-source home energy management tool used as a secondary baseline, called through its REST interface
Data processing	pandas 3.0, NumPy 2.5	REFIT cycle extraction, price series and scenario construction
Tracing	LangSmith	recording the reasoning trace and tool calls of every run
Statistics	SciPy 1.18	Wilcoxon signed-rank test, TOST equivalence test, effect size and Spearman correlation

5. Evaluation and Results

This chapter presents the evaluation of the artifact, corresponding to DSR Phases 4 and 5. Section 5.1 describes the benchmark: the appliance and price data and the construction of the 48 evaluation scenarios across six price-volatility types. Section 5.2 uses the MILP solver alone to examine Finland's move to fifteen-minute electricity prices, quantifying how much of the achievable scheduling saving depends on fifteen-minute rather than hourly price information; this establishes where differences between schedulers can be expected before any agent is evaluated. Section 5.3 reports the demonstration of the pipeline on representative scenarios (DSR Phase 4). Sections 5.4 and 5.5 compare the optimality gaps of the multi-agent pipeline and the single-agent baseline across all scenarios and test the difference with the Wilcoxon signed-rank test. Section 5.6 compares both designs with two simpler baselines, a rule-based smart timer and a language model without tools, and with two real home energy management tools, the EMHASS optimizer and a Home Assistant cheapest-hours automation. Section 5.7 repeats the tests of H1 and H2 on the held-out fifteen-minute benchmark, as fixed in the pre-registration. Section 5.8 breaks the cost savings down by appliance type. The agent results in Sections 5.3–5.8 are pending the full evaluation runs.

5.1 Dataset and scenario design

The evaluation benchmark combines real appliance consumption profiles from UK households with real Finnish day-ahead electricity prices. Each of the 48 scenarios describes one day in 96 fifteen-minute slots (00:00–24:00 local time) and consists of a price curve, a household base load, a set of flexible appliance cycles with user time windows, and a household peak load limit. The benchmark is generated by a deterministic, seeded build procedure, so it can be reproduced exactly from the source data.

5.1.1 Appliance data

Appliance profiles are drawn from the cleaned REFIT Electrical Load Measurements dataset (Murray et al., 2017; Murray & Stankovic, 2016), which records the whole-house aggregate and nine individually monitored appliances at 8-second intervals in 20 UK households between 2013 and 2015. The cleaned release corrects timestamps for daylight saving time and re-aligns appliance monitor columns where a monitor was moved between appliances during the study. The appliance-to-column mapping follows the dataset documentation.

Eighteen households are used. House 12 is excluded because it has no monitored washing machine, dishwasher or tumble dryer. House 19 is excluded after inspection: its washing-machine monitor recorded no heating phase in any of its 246 detected cycles (no

fifteen-minute mean above 1.2 kW, against 22–70 % of cycles in the other households), so its profiles do not represent the water-heating load that dominates the cost and peak demand of a wash. Washer-dryers (houses 1, 8, 9 and 18) are not used as profiles, because they run washing and drying as a single cycle; their consumption is nevertheless removed from the base load.

Individual appliance cycles are extracted from each monitor by thresholding the one-minute mean power at 20 W and merging pauses shorter than 30 minutes for dishwashers (whose programmes contain a passive drying pause) and 15 minutes for washing machines and tumble dryers (which are often reloaded within minutes). Cycles shorter than 20 minutes, longer than 300 minutes or below 0.1 kWh are discarded, and each remaining cycle is converted into a fifteen-minute mean power profile. This yields 6,172 washing-machine cycles from 19 monitored machines, 4,702 dishwasher cycles from 15 machines and 1,940 tumble-dryer cycles from 7 machines. To prevent heavily used appliances from dominating the benchmark, a scenario first draws a monitored appliance of the required type and then one of its cycles. The household base load is the median, per fifteen-minute slot of the day, of the aggregate minus all shiftable appliance monitors, computed after discarding readings that the dataset flags as inconsistent (sub-metered sum exceeding the aggregate).

5.1.2 Electricity price data

Day-ahead spot prices for the Nord Pool FI bidding zone were obtained for 1 January 2022 to 24 September 2026 from the public Sähkötin price service, which republishes Nord Pool data (Pakastin Oy, 2026). The values are in EUR/MWh excluding VAT and were verified against an independent source for the December 2022 crisis period. The service provides hourly values, which are repeated across the four fifteen-minute slots of each hour; days with a daylight-saving transition are dropped, leaving 1,718 complete days.

Scheduling cost is computed with the consumer price rather than the raw spot price. For spot price s, the consumer price per kWh is

$$p = (s + m + d + \tau)(1 + v),$$

with a retail margin $m = 0.5$ c/kWh, a distribution fee $d = 5.0$ c/kWh, the electricity tax $\tau = 2.253$ c/kWh and VAT $v = 25.5\,\%$. These are representative values for a Finnish household on a spot-price contract; a single VAT rate is applied to all days so that scenarios from different periods remain comparable. Using the consumer price keeps costs positive on days with negative spot prices, which keeps the optimality gap defined in Section 3.2 meaningful. Only the energy of the flexible appliances is costed, since the base load costs the same under every schedule.

5.1.3 Scenario construction

Table 5.1 summarises the six scenario types. Days are assigned to types by the within-day standard deviation of the spot price. Types A and B use disjoint days from the lowest quartile among days without negative prices, type C uses days between the 40th and 60th percentiles, and type D uses days in the top 15 % that fall in December–February. Type E uses days from December 2022, sampled from the 16 days with the highest maximum price. Type F uses days with at least two hours of negative prices. Eight days are sampled per type.

Types C–F need up to five loads, while REFIT provides three flexible appliance types; a scenario may therefore contain two cycles of the same type, such as two washing loads. Each appliance has a user window (earliest start and deadline), and its naive start time is the earliest start. Types A–C use windows opening between 06:00 and 12:00 and lasting 8–14 hours. In type D all five appliances share one window from 08:00–10:00 until 22:00–24:00, so their cheapest start times coincide. In type E each appliance becomes ready 0–2 hours before the day's highest daytime price and has 4–8 hours of slack beyond its cycle, so the naive schedule runs into the price spike while avoiding it remains possible. This reflects that price spikes coincide with evening demand peaks, when households tend to start appliances. Type F windows open between 00:00 and 04:00 and last until midnight.

The household peak load limit, including the base load, is 3.5 kW. It represents a household power cap, for example one induced by a power-based distribution tariff, rather than the main fuse (typically 3×25 A, about 17 kW), which would never bind for these appliances. The build procedure verifies with the MILP solver that every scenario has a feasible schedule; this held for all 48 scenarios without relaxing the limit. Table 5.2 reports the resulting characteristics per type. A scenario requires coordination when running every appliance at its individually cheapest start time exceeds the peak limit; these 28 scenarios form the basis of the conflict resolution rate.

Table 5.1 Evaluation scenario design.

Type	Price regime	Appliances	Inst.	Research purpose
A	Stable, low volatility	2 appliances: dishwasher + washing machine	8	Baseline check: both systems should perform well
B	Stable, low volatility	3 appliances: dishwasher + washing machine + tumble dryer	8	Tests three-way conflict resolution
C	Moderate volatility	3 to 5 appliances mixed	8	Tests price-signal responsiveness
D	High volatility, winter peak	5 appliances, overlapping constraints	8	Most financially consequential condition
E	Price spike (Dec 2022 crisis)	3 appliances, ready 0–2 h before the daily price peak, 4–8 h slack	8	Extreme Finnish price event robustness
F	Negative spot price (surplus)	4 appliances, flexible windows	8	Tests use of low-cost windows on negative spot-price days

Table 5.2 Benchmark characteristics per scenario type.

Type	Price days	Mean appliances	Require coordination	Naive schedule within limit	MILP saving vs. naive
A	2022–2026	2.0	1/8	8/8	10.2 %
B	2023–2026	3.0	4/8	7/8	4.3 %
C	2023–2026	3.4	5/8	6/8	19.9 %
D	2022–2026 (Dec–Feb)	5.0	8/8	0/8	37.0 %
E	1–14 Dec 2022	3.0	4/8	3/8	14.4 %
F	2023–2025	4.0	6/8	5/8	7.8 %
All		3.4	28/48	29/48	

5.2 Value of 15-minute price information

Since 1 October 2025 the Nord Pool day-ahead market settles Finnish electricity in fifteen-minute periods. The main benchmark uses hourly prices, because its type E days date from the 2022 energy crisis. To examine the new market design, a second set of 48 scenarios (Q-A01 to Q-F08) was built with the same procedure from native fifteen-minute FI prices for 1 October 2025 to 25 September 2026 (358 complete days), obtained from the public price service of the Estonian transmission system operator, which republishes Nord Pool data (Elering AS, 2026). Its hourly averages match the hourly series used in Section 5.1. Because this period contains no crisis month, its type E days are drawn from the 16 days with the highest peak price (up to 65 c/kWh), and no day is used by more than

one type. Prices genuinely vary within the hour: in 98 % of hours the four quarter-hour prices differ, with a median spread of 0.75 c/kWh.

For each Q scenario, two optimal schedules were computed with the MILP solver. The first is planned on the real fifteen-minute prices. The second is planned on hourly averages, which is all the price information a household had before the switch, and is then settled at the real fifteen-minute prices. Hourly prices make the four quarters of an hour tie, so ties are broken towards the earliest start. The cost difference between the two schedules is the saving that is only reachable with fifteen-minute price information. Table 5.3 reports the savings of both schedules relative to the naive schedule, the additional cost reduction from fifteen-minute information, and the share of the achievable saving that requires it; the "All" row weights scenarios by their cost.

Table 5.3 Optimal schedule cost with hourly vs. 15-minute price information (MILP, Q scenarios).

Type	Intra-hour spread (c/kWh)	Saving, hourly plan	Saving, 15-min plan	Additional cost reduction	Share of saving needing 15-min	Scenarios improved
A	0.41	4.1 %	4.2 %	0.1 %	2.4 %	5/8
B	0.33	3.2 %	3.4 %	0.3 %	6.4 %	8/8
C	1.37	7.1 %	8.6 %	1.7 %	16.4 %	8/8
D	4.96	27.6 %	28.6 %	1.4 %	3.5 %	8/8
E	4.91	35.4 %	37.0 %	2.1 %	2.8 %	6/8
F	0.21	4.6 %	4.7 %	0.1 %	1.7 %	4/8
All	2.03	23.7 %	24.7 %	1.2 %	3.7 %	39/48

Fifteen-minute price information lowers the optimal cost in 39 of the 48 scenarios, but the effect is modest: 1.2 % overall, or 3.7 % of the achievable saving. The remaining 96 % of the saving is available to a scheduler that plans on hourly prices, since most of the value lies in moving appliance cycles between hours rather than between quarters of an hour. The additional reduction grows with the intra-hour price spread (Spearman $\rho = 0.66$ across scenarios) and is largest on volatile days: 1.4 % in type D, 2.1 % in type E, and 10.2 % in the single largest case (Q-E03, 17 October 2025). Type C shows the largest relative share (16.4 %), because its hourly price differences are small, so intra-hour variation forms a larger part of the opportunity. In absolute terms the gain is about one cent per household per day for the two to five cycles scheduled, and about three cents on type D days. The schedules nevertheless differ: 110 of the 163 appliance start times change between the two plans. These results indicate that the hourly main benchmark captures most of the scheduling problem, while any difference between the agents under fifteen-minute prices is expected to appear mainly in types C, D and E.

5.3 Demonstration results

Table 5.4 shows what each system recommends for one day, scenario C03 (28 September 2023). The household has four appliance cycles: a 135-minute tumble dryer cycle ready from 06:00, a second 45-minute dryer cycle ready from 08:45, a dishwasher ready from 08:00 and a washing machine ready from 11:45, each with a deadline in the evening, and a peak load limit of 3.5 kW. Starting every cycle as soon as it is ready would cost €0.78.

Table 5.4 Daily recommendations for scenario C03 (28 September 2023), run 1 with GPT-4o-mini.

System	Dryer 1	Dryer 2	Dish-washer	Washing machine	Saving vs. no scheduling
No scheduling	06:00	08:45	08:00	11:45	–
MILP optimum	06:00	16:15	15:00	15:30	€0.139
EMHASS, profiles	06:00	16:15	15:00	15:30	€0.139
Smart timer	06:00	16:00	15:00	16:00	€0.139
Multi-agent pipeline	06:00	16:00	15:00	16:00	€0.139
Single agent	06:00	16:00	15:00	15:15	€0.137
Cheapest hours	06:00	15:00	08:00	14:00	€0.080
LLM without tools	06:00	08:45	08:00	11:45	€0.000

The pipeline, the single agent, the smart timer and EMHASS all move the dishwasher, the second dryer and the washing machine into the cheap afternoon hours and capture almost the full saving of €0.14. The cheapest-hours automation moves the second dryer and the washing machine but leaves the dishwasher at its ready time: the blueprint starts a cycle at the first slot whose price is at or below its threshold, and the price at 08:00 already qualifies, although cheaper hours follow later. The language models also explain their recommendation. The pipeline's Orchestrator wrote: "The first tumble dryer will start at 06:00, followed by the dishwasher at 15:00, and the second tumble dryer and washing machine will both run at 16:00 … staying within your peak load limit of 3.50 kW." The explanation of the LLM without tools illustrates the risk of an ungrounded message: it states that the schedule "was optimized to run the appliances during the lowest price slots", while its start times are exactly those of no scheduling and save nothing. The pipeline's summary is checked against the final peak-load test before it is written (Section 4.2), whereas nothing checks the claims of the LLM without tools.

The design iterations that led to the evaluated version of the pipeline are described in a later revision of this section.

5.4 Quantitative results

Work in progress.

5.5 Statistical analysis

This section reports the tests described in Section 3.2 for GPT-4o-mini on the main benchmark, with the mean over three runs per scenario as the paired observation ($N = 48$). These results are exploratory; the pre-registered confirmatory results are reported in Section 5.7. The results for gpt-oss-20b are added when its runs are complete.

Schedule quality. The multi-agent pipeline reached a mean optimality gap of 1.25 % and the single agent 1.43 %. The Wilcoxon signed-rank test finds no significant difference ($W = 449$, $Z = -0.14$, $p = 0.89$, $r = 0.02$), and the median paired difference is zero, because both designs reach the same cost in most scenarios. The equivalence test shows that this is not merely a lack of evidence. The mean paired difference is -0.18 percentage points, with a 90 % confidence interval of $[-0.71, 0.35]$, which lies inside the primary margin of ± 1 percentage point (Table 5.5). The two designs are therefore equivalent in schedule quality within one percentage point of the optimum. At the narrower margin of 0.5 percentage points the t version of the test does not conclude equivalence, because a few scenarios with large differences in either direction widen the interval, although the Wilcoxon version does.

Table 5.5 Equivalence test (TOST) of the optimality gap, multi-agent vs. single-agent (GPT-4o-mini, $N = 48$).

Margin	p, paired t TOST	p, Wilcoxon TOST	Equivalent
± 0.5 percentage points	0.156	< 0.001	No
± 1 percentage point (primary)	0.006	< 0.001	Yes
± 2 percentage points	< 0.001	< 0.001	Yes

Efficiency. The two designs differ significantly in every efficiency measure (Table 5.6). The pipeline used 34 % fewer input tokens, because each specialist works with a short context and only its own tools, while the single agent carries the entire conversation and all eight tool descriptions through every step. It produced 83 % more output tokens and made 23 % more tool calls, since four agents each write their own structured output and call their tools separately. Input tokens dominate the price, so the API cost per run was 21 % lower for the pipeline, and lower in 35 of the 48 scenarios. The sequential hand-offs made the pipeline 69 % slower, with a mean of 21.4 seconds against 12.7 seconds, and it was faster in only two scenarios.

Taken together, decomposing the task into specialist agents did not lower the cost of the schedules, but it reached the same quality with about a third fewer tokens and a fifth lower inference cost, at the price of a longer response time.

Table 5.6 Efficiency per run, multi-agent vs. single-agent (GPT-4o-mini; tokens and tool calls per run, API cost in USD, latency in seconds; each value is the mean over $N = 48$ scenarios of the per-scenario mean of three runs; Wilcoxon signed-rank test).

Measure	Single agent	Multi-agent	Difference	Multi-agent lower	p	r
Input tokens	21 686	14 292	−34 %	43/48	< 0.001	0.79
Output tokens	661	1 208	+83 %	1/48	< 0.001	0.87
Total tokens	22 347	15 500	−31 %	40/48	< 0.001	0.75
API cost (USD)	0.0037	0.0029	−21 %	35/48	< 0.001	0.54
Latency (s)	12.7	21.4	+69 %	2/48	< 0.001	0.86
Tool calls	16.5	20.3	+23 %	1/48	< 0.001	0.81

5.6 Comparison with baselines and real tools

Table 5.7 compares the two agent designs with the smart timer, the LLM without tools and the two real home energy management tools. All systems that use a language model use GPT-4o-mini.

Table 5.7 Agent designs, simpler baselines and real home energy management tools (GPT-4o-mini, main benchmark, 48 scenarios).

System	Feasible runs	Mean gap	Median gap	Gap, type D	Savings captured	Time per run
Smart timer	48/48	0.23 %	0.00 %	1.2 %	99 %	< 0.01 s
EMHASS, profiles	48/48	0.32 %	0.00 %	0.1 %	99 %	1.2 s
Multi-agent pipeline	144/144	1.25 %	0.01 %	5.9 %	93 %	21.4 s
Single agent	143/144	1.43 %	0.00 %	6.5 %	90 %	12.7 s
EMHASS, nominal power	48/48	2.24 %	0.45 %	8.5 %	87 %	1.0 s
Cheapest hours	25/48	17.6 %	4.2 %	62.7 %	25 %	< 0.01 s
LLM without tools	126/144	23.1 %	11.9 %	66.0 %	13 %	2.0 s

Without tools, GPT-4o-mini schedules poorly. Its mean optimality gap is 23.1 %, and 18 of its 144 schedules are infeasible: 14 exceed the peak limit and 4 start a cycle outside its window. A further 22 of its feasible schedules cost more than starting every appliance as soon as it is ready, and over all its runs it captures only 13 % of the achievable saving, against about 90 % for the agents. Both agent designs are significantly better (Wilcoxon $p < 0.001$, $r = 0.87$ for each), with median paired differences of 10.8 percentage points for the single agent and 11.4 for the pipeline. The same model that fails to schedule from a single prompt comes close to the optimum when it can query prices, costs and loads through tools and check its own work. Tool use, rather than the language model's own calculation, is therefore what makes the agents viable.

The smart timer, however, outperforms both agent designs. It finds a feasible schedule in every scenario, is optimal in 30 of the 48, and its largest gap is 3.7 % (D04). Both agent

designs have significantly larger gaps than the timer (single agent $p < 0.001$, $r = 0.63$; pipeline $p < 0.001$, $r = 0.68$). The difference is concentrated in type D, where the timer's mean gap is 1.2 % against 5.9 % for the pipeline and 6.5 % for the single agent. The median paired differences are small (0.002 percentage points for the single agent and 0.017 for the pipeline), because in most scenarios all three systems reach the optimum or come very close to it. The timer needs no language model, runs in about two milliseconds and has no inference cost. In this benchmark, where the power profile and window of every appliance are known exactly and the horizon is one day, placing the appliances one at a time is almost always enough. The agents come within about one percentage point of this rule but do not improve on it. What they add is the ability to start from a request written in natural language rather than from structured appliance data, and to explain the schedule they produce.

The two real home energy management tools, added after the pre-registration, bracket the agents. Configured with the measured power profiles, EMHASS comes close to the optimum: every schedule is feasible, 32 of the 48 are optimal, and the mean gap is 0.32 %. Its largest gap, 4.2 % in F02, comes from its default solver tolerance of 1 % (the setting `lp_solver_mip_rel_gap` = 0.01; EMHASS contributors, 2026), which applies to the whole electricity bill including the base load and can therefore amount to several percent of the appliance cost scored here. With an exact tolerance the same optimization comes within 0.2 % of the optimum in every scenario, which confirms that the MILP reference reflects what a deployed open-source optimizer can achieve. The agents' mean gaps are about one percentage point higher, but the difference from EMHASS as deployed is not significant (Wilcoxon $p = 0.24$ for the single agent and $p = 0.19$ for the pipeline). Configured with a constant nominal power, EMHASS still keeps every schedule within the peak limit, because it plans each cycle at its peak power, but it misjudges the cost of the real cycles and reaches a mean gap of 2.2 %. Both agent designs are significantly better than this configuration (Wilcoxon $p < 0.001$ for each, $r = 0.67$ for the single agent and 0.64 for the pipeline). The cheapest-hours automation, under the assumption stated in Section 3.2 that its trigger starts a cycle which then runs to completion, exceeds the peak limit in 23 of the 48 scenarios, because each appliance follows its own cheap period and appliances whose windows share a price minimum start together. Scored at the naive cost when infeasible, it reaches a mean gap of 17.6 %, and both agent designs are significantly better ($p < 0.001$, $r = 0.82$ and 0.81). The agents are therefore outperformed by the smart timer and are not significantly different from a well-configured optimizer as deployed, while they outperform a simply configured optimizer and the per-device automation that is common among households.

Two checks show that this ranking does not depend on how the costs are scored. First, scoring an infeasible run at the cost of the naive schedule does not drive the result. Restricted to feasible runs, the mean gaps are 0.23 % for the smart timer, 0.32 % for EMHASS with profiles, 1.25 % for the pipeline, 1.45 % for the single agent, 2.24 % for EMHASS with nominal power, 7.9 % for the cheapest-hours automation and 23.0 % for the LLM without tools, the same order as in Table 5.7, and the order on the held-out benchmark is also unchanged. The cheapest-hours automation looks better on feasible runs only because these are the days on which its appliances happen not to collide.

Second, the ranking is the same under the spot price alone as under the consumer price, as explained in Section 3.2. The Discussion returns to what these results imply for the artifact.

5.7 Confirmatory evaluation

The hypotheses of SQ1 and SQ2 were fixed in the pre-registration, which was committed to the project repository on 30 September 2026 together with the code, prompts, tools and analysis script. The confirmatory run followed on the held-out benchmark of 48 scenarios with fifteen-minute prices (Section 5.2), on which no language model had been run before. All systems that use a language model used GPT-4o-mini with three runs per scenario, and the analysis was run exactly as registered. The run cost USD 0.92 in API fees. Table 5.8 reports the six registered tests.

Table 5.8 Pre-registered confirmatory tests on the held-out fifteen-minute benchmark (GPT-4o-mini, $N = 48$; p-values Holm-adjusted within each sub-question).

	Hypothesis	Estimate	p	r	Supported
H1	Single agent gap < LLM without tools	-17.9 pp	< 0.001	0.86	Yes
H1	Pipeline gap < LLM without tools	-17.7 pp	< 0.001	0.86	Yes
H2a	Pipeline equivalent to single agent (± 1 pp)	$+0.22$ pp	< 0.001	–	Yes
H2b	Pipeline uses fewer input tokens	-31 %	< 0.001	0.84	Yes
H2b	Pipeline has a lower API cost per run	-21 %	< 0.001	0.70	Yes
H2c	Pipeline has a longer latency	$+52$ %	< 0.001	0.80	Yes

All six hypotheses are supported. For H1, both agent designs reached a mean optimality gap of about one percent (0.86 % for the single agent and 1.08 % for the pipeline), whereas the LLM without tools reached 18.8 %. Fifteen of its 144 schedules were infeasible, and over all its runs it captured only 24 % of the achievable saving, against 92 % for the single agent and 90 % for the pipeline. For H2a, the mean paired difference in gap was $+0.22$ percentage points with a 90 % confidence interval of $[-0.15, 0.59]$, which lies well inside the margin of ± 1 percentage point. On this benchmark the pipeline was thus marginally worse than the single agent on average, the reverse of the exploratory result, but in both runs the difference is small enough to count as equivalent. For H2b and H2c, the pipeline used 13 310 input tokens per run against 19 189 for the single agent, its API cost was 21 % lower, and its latency was 20.4 seconds against 13.4 seconds.

The confirmatory estimates are close to the exploratory ones: input tokens fell by 31 % in both runs, API cost by 21 % in both, and latency rose by 52 % here against 69 % before. The secondary results also replicate. The equivalence test is again not significant at the narrower margin of ± 0.5 percentage points in its t version ($p = 0.11$), although its Wilcoxon version is. The smart timer again outperforms both agent designs, with a mean gap of 0.17 % (Wilcoxon $p < 0.001$ against each design, $r = 0.69$ for the single agent and 0.64 for the pipeline). Type D remains the hardest scenario type, with mean gaps of

3.9 % for the single agent and 5.5 % for the pipeline, while the LLM without tools fails most clearly in type E, with a mean gap of 56 %.

The real home energy management tools were added after the pre-registration, so the following comparisons are secondary and not registered; their p-values are descriptive and not adjusted for multiple testing. EMHASS as deployed reaches a mean gap of 0.36 % with measured profiles, not significantly different from either agent design (Wilcoxon $p = 0.90$ for the single agent and $p = 0.69$ for the pipeline), and 3.5 % with nominal power, significantly worse than both ($p < 0.001$). The cheapest-hours automation exceeds the peak limit in 26 of the 48 scenarios, with a mean gap of 18.7 %. These patterns match those on the main benchmark.

The registered findings, H1 for SQ1 and H2 for SQ2, are therefore not specific to the benchmark on which the pipeline was developed: they hold on unseen days with a different price resolution. They cover GPT-4o-mini only. The part of SQ2 that concerns a small model run locally (H3) is tested on the main benchmark and reported separately.

5.8 Per-appliance breakdown

Work in progress.

6. Discussion

Work in progress. This chapter will interpret the results in relation to the research questions, connect findings to the agentic AI trends reviewed in Chapter 2, and address the limitations and directions for future research.

6.1 Interpretation of research questions

Work in progress.

6.2 Agents compared with simpler schedulers

The comparison with simpler baselines in Section 5.6 places the results in context, and two findings stand out. Both were first observed on the main benchmark and then replicated on the held-out fifteen-minute benchmark (Section 5.7), where the first was confirmed as a pre-registered hypothesis. First, tool use is essential. The same language model that captures about 90 % of the achievable saving as an agent captures only 13 to 24 % when it has to work out the schedule in a single answer, and 10 to 13 % of those schedules break the peak limit or a time window. This agrees with the motivation for ReAct (Yao et al., 2023), in which reasoning is grounded in the observations returned by actions instead of being carried out unaided. It indicates that the role of the language model in the pipeline is to decide which calculations to request and how to combine their results, not to perform the calculations itself.

Second, on well-specified single-day problems a rule-based smart timer is slightly better than either agent design and costs almost nothing to run. This held on both benchmarks, with the timer's mean gap at 0.23 % and 0.17 % against about one percent for the agents. The comparison with real tools adds two observations. A deployed optimizer, EMHASS, comes close to the optimum when it is given each appliance's measured power profile, and with its default settings it is not significantly better than the agents, but it falls behind them when it is configured with a constant nominal power. The quality of an optimizer thus depends on how well its configuration describes the appliances. The per-device cheapest-hours automation that is common among households exceeds the peak limit on about half of the days, which shows that coordinating appliances, and not only finding cheap hours, is where simple automation fails. For a household whose appliances and time windows are already known to a home energy management system in structured form, the evidence therefore does not support replacing such a rule with a language model agent on cost grounds. The case for the agents rests on what surrounds the schedule instead. Their recommendation comes with a plain-language explanation, they check it against the household data, and in principle they could take the household's wishes in

everyday language rather than as structured settings, while keeping the cost within about one percentage point of the rule. This study does not test that last ability, because the daily appliance plan is given in structured form. Plans stated informally, incompletely or inconsistently, with a measure of how accurately each system turns them into constraints, would be the most direct way to establish where the agents add value.

6.3 Limitations

Several limitations of this study warrant acknowledgement. First, the evaluation is entirely simulation-based using historical data. While the use of real consumption profiles and real Nord Pool FI prices provides stronger ecological validity than synthetic scenarios, the study does not evaluate the pipeline in a live deployment where appliance states change dynamically and user preferences may deviate from scripted constraint windows. Second, the REFIT dataset was collected from UK homes between 2013 and 2015 and may not fully reflect the appliance configurations and usage patterns of contemporary Finnish households. Third, LLM outputs are non-deterministic: two runs of the same scenario may produce different schedules. To mitigate this, each scenario is run three times and results are averaged, but residual variance from non-determinism remains. Fourth, the study evaluates only three appliance types; generalizing findings to electric vehicle chargers, heat pumps, or battery storage requires additional evaluation. Fifth, the daily appliance plan is given in structured form, with an explicit window for every appliance. The study therefore does not test the agents' ability to interpret wishes stated informally, which is where their advantage over rule-based scheduling is expected, and on these well-specified problems a rule-based smart timer outperforms both agent designs (Section 5.6).

6.4 Future research

Several directions for future research emerge from this study. First, extending the pipeline to thermal loads such as electric space heating, heat pumps and domestic hot water systems would address the gap explicitly identified by El Makroum et al. (2025). Second, implementing the price API and data reader as MCP servers would enable reuse by future researchers without custom integration code, contributing to the emerging standardization of agent tool interfaces. Third, a live deployment study involving Finnish households on spot-price contracts would provide ecological validity that simulation-based evaluation cannot. Fourth, extending the multi-LLM comparison to include Qwen-3-32B and Phi-4 would provide a more complete picture of the SLM landscape for residential scheduling tasks. Fifth, a benchmark of informal household requests with annotated constraints would test the agents where a rule-based scheduler cannot operate, by measuring how accurately each system turns such requests into a correct schedule.

7. Conclusions

Work in progress. This chapter will summarize the main contributions of the thesis, explicitly answer the RQ, SQ1 and SQ2 using evidence from Chapter 5, and state the implications for academic research on multi-agent LLM systems and practical deployment of intelligent demand flexibility tools in Finnish residential settings.

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